

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

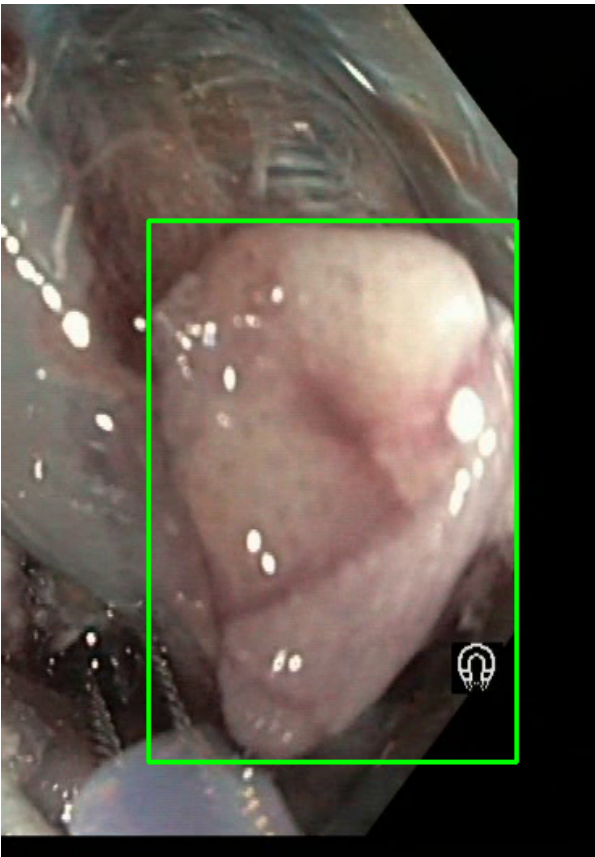
Video ID:	52cf5162-bdcb-42dc-8ce9-e946c899392c
Total Frames:	15004
Frame Rate:	25.0 fps (assumed)
Duration:	600.2 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	2251	✓
RT-DETR Detections	6935	✓
MedSAM2 Detections	6113	✓
Consensus (3-Model Agreement)	2	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
MEDIUM RISK	1	50.0%
LOW RISK	1	50.0%
TOTAL ANALYZED	2	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 390 frames)



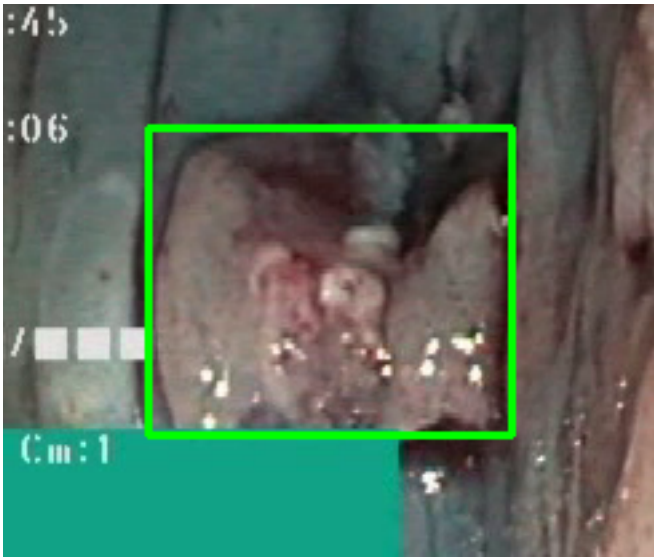
Feature	Value	Interpretation
Frame Index	13568	Frame 13568 of 15004
Detection Method	Detector Consensus (MedSAM2, UDETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 72.5%
Redness Score	0.070	Color-based vascularization
Relative Radius	25.2%	Polyp size as % of frame diagonal
Texture Score	0.266	Surface morphology
Vessel Visibility	0.140	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket

Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	83.1%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #2 — LIFTED_POLYP [LOW_RISK] (Tracked across 25 frames)



Feature	Value	Interpretation
Frame Index	12094	Frame 12094 of 15004
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 68.5%
Redness Score	0.060	Color-based vascularization
Relative Radius	17.8%	Polyp size as % of frame diagonal
Texture Score	0.284	Surface morphology
Vessel Visibility	0.013	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	77.5%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

CLINICAL IMPRESSION

Summary:

A total of 2 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 1 (50.0%)
- LOW RISK polyps: 1 (50.0%)

Recommendation:

The presence of MEDIUM RISK polyps warrants clinical review and correlation with endoscopic findings.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.